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Name Ramith. H.

Standard Section 5T Roll No.

Subject Computer Networks (CN Lab)

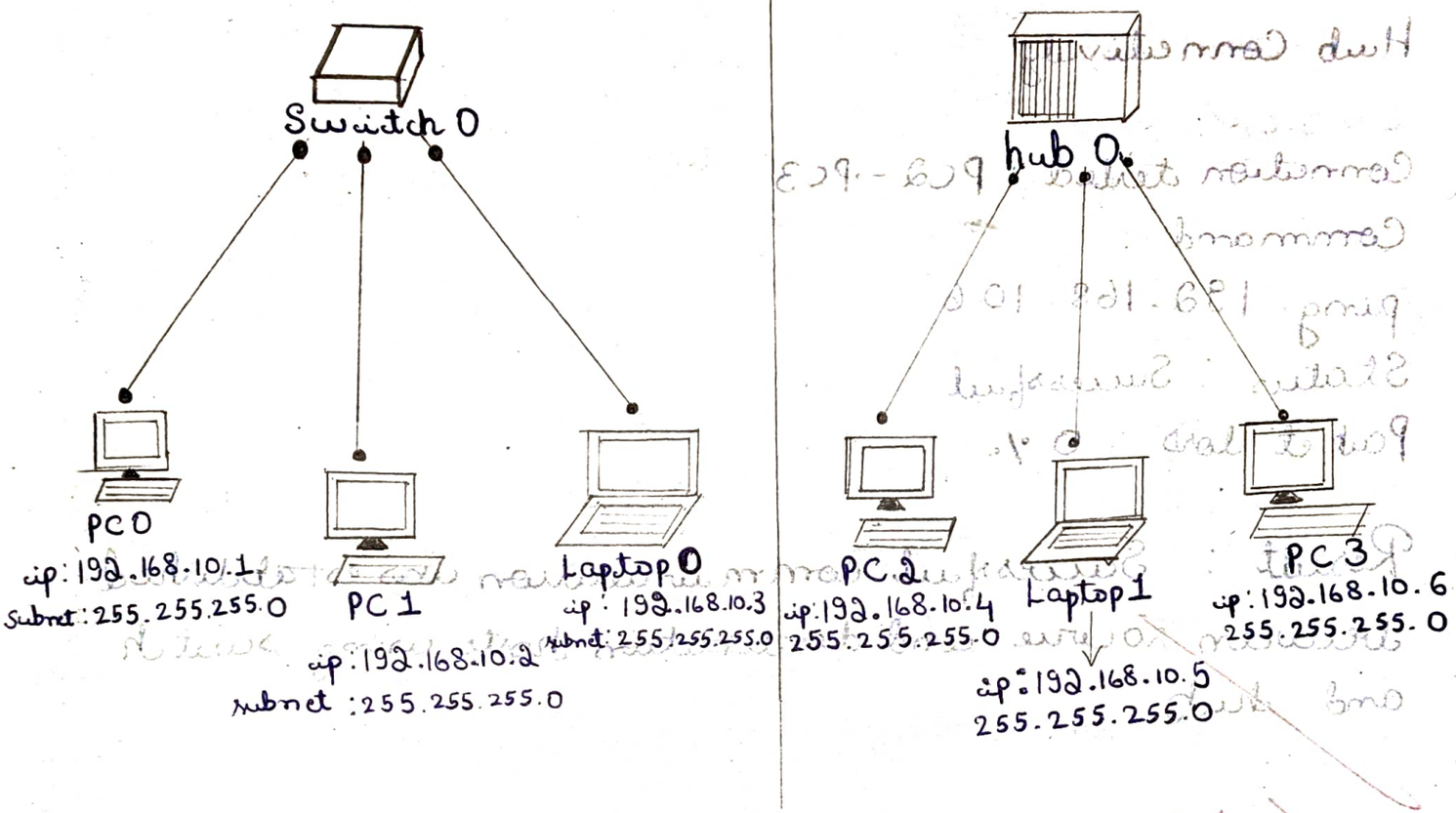
SL No.	Date	Title	Page No.	Teacher Sign / Remarks
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Cycle - 1

1> Create a topology and simulate sending a simple PDU from source to destination using hub and switch as connecting devices and demonstrate ping message

Step 1 : Create a simple topology



Step 2: Configure a Static IP address and subnet mask for each host. Example : 192.168.10.1 . subnet mask : 255.255.255.0

Step 3 : Repeat ip configuration for all hosts in topology

Step 4 : Select simple PDU from right side of packet tracker

Step 5 : Select the source PC and then the destination PC to test connectivity

Step 6 : Open Desktop -> Command Prompt on source PC and use ping command : ping <destination IP address>

Eg: PC0 to PC1 : ping 192.168.10.2

Step 7 : Observe PDU result and ping response

working :

- Switch : Forwards frame to required destination port
- Hub : Broadcasts the incoming frame to all connections

Output :

Switch Connectivity

Connection tested : PC0 -> PC1

Command :

ping 192.168.10.2

Status : Successful

Packet loss : 0 %.

Hub Connectivity...

Connection tested : PC2 -> PC3

Command :

ping 192.168.10.6

Status : Successful

Packet loss : 0 %.

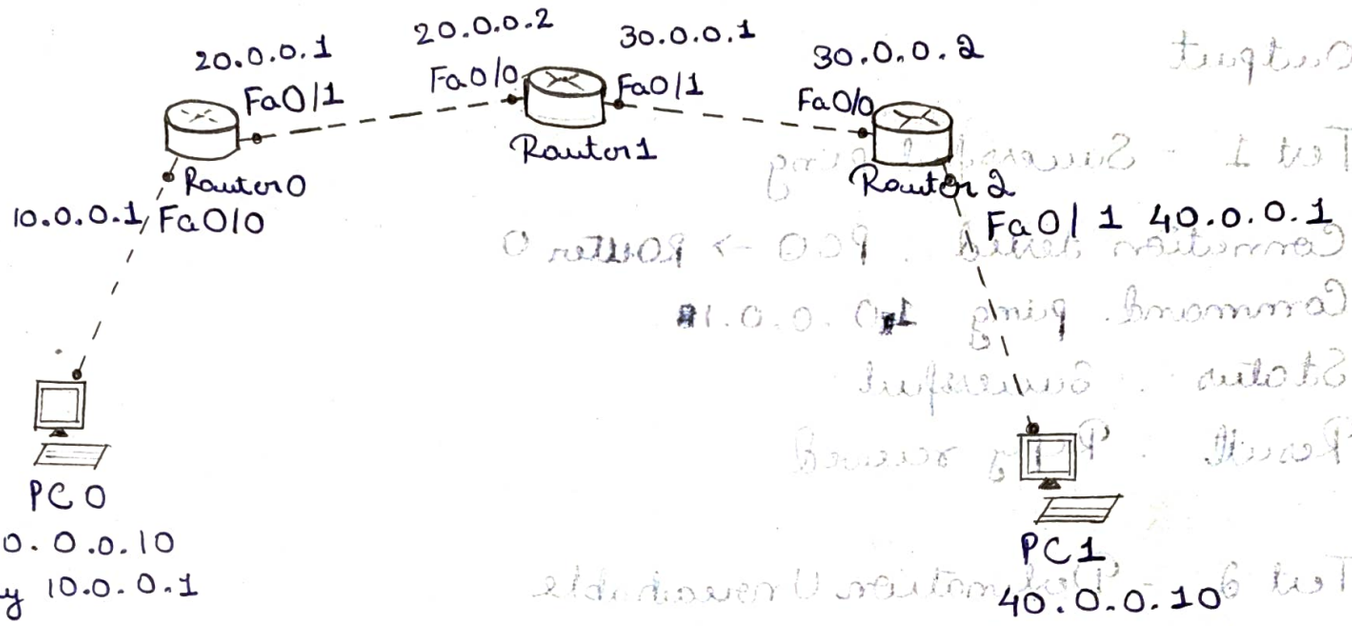
Result : Successful communication was established between source and destination hosts using switch and hub

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2> Configure IP adress to routers in packet Tracer. Explore following ping, messages, responses, destination unreachable, request timed out, reply

Step 1 : Create a topology like this. Configure IP adress and gateway to hosts



Step 2 : Configure IP to Router

Open R1 → CLI

Continue with configuration dialog? [yes/no] : no

Router > enable

Router #configure terminal

Router(config)# interface fa0/0

Router (config-if)# ip address 10.0.0.1 255.255.255.0

Router (config-if)# no shut down

Router (config-if)# exit

Router (config)# interface fa0/1

Router (config-if)# ip address 20.0.0.1 255.255.255.0

Router (config-if)# no shutdown

Router (config-if)# exit, #exit

·Click on Router R1, then click CLI, give no and enter, now you will go user mode and enable press enter, you get into Privileged mode, type configure terminal and enter to get into global configuration mode.

Step 3 : Now configure router interface with ip and subnet mask then give no shutdown to make this interface and line protocol up

Step 4 : Now lights on all ports become green from red. Now open cmd of PC and check for Ping messages, destination reachable, request time out and Reply

Step 5 : Test Connectivity

Output

Test 1 - Successful ping

Connection tested : PC0 → Router 0

Command : ping 10.0.0.1

Status : Successful

Result : Reply received

Test 2 - Destination Unreachable

Connection tested : PC0 → 40.0.0.10 (R2)

Command : ping 40.0.0.10

Status : Destination host unreachable

Test 3 - Requested time out

Connection tested : PC0 → 10.0.0.6

Command : ping 10.0.0.6

Status : Requested time out

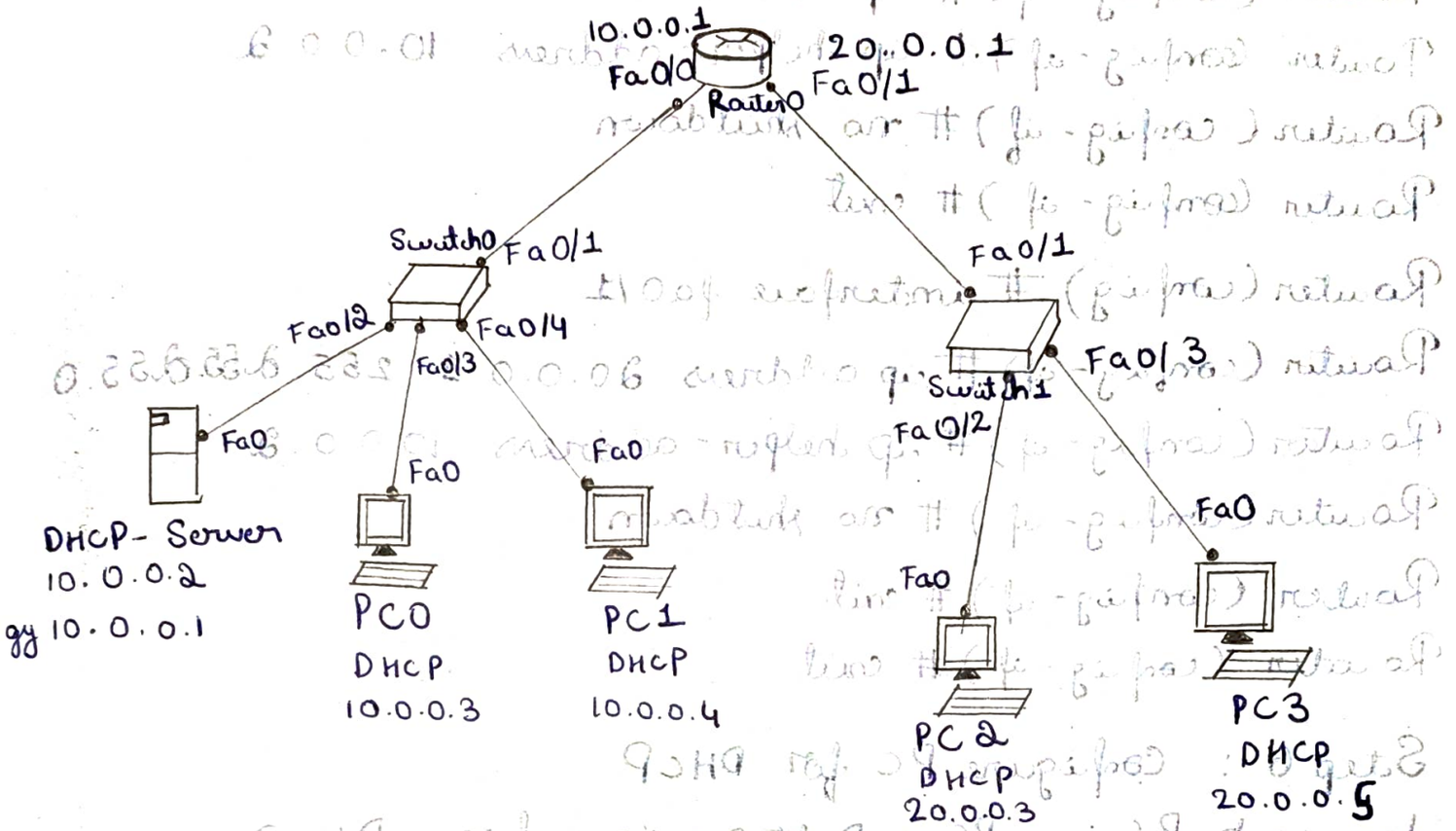
Requested time out

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3> Configure DHCP within inside LAN and outside LAN

Step 1: Create a LAN like this



- Step 2: Configure DHCP server Gateway

  DHCP Server → Config → Settings

  Select static under Gateway/DNS and enter

  Gateway: 10.0.0.1

- Step 3: Configure DHCP server IP

  DHCP Server → Config → Fast Ethernet 0

  Set: IP adress: 10.0.0.2

  Subnet: 255.0.0.0

- Step 4: Create DHCP Pools

  Go to:

  DHCP Server → Services → DHCP

  Turn DHCP: ON

  Create pools for two networks

  Pool: Switch1

  Default Gy: 10.0.0.1 (20)

  Start ip: 10.0.0.3 (20)

  Subnd mask: 255.0.0.0

  Max no users: 20

  Similarly for second pool

Step 5 : Configure Router

Open Router 0 -> cli

Router > enable

Router # configure terminal

Router (config) # interface fa0/0

Router (config-if) # ip address 10.0.0.1 255.255.255.0

Router (config-if) # ip helper-address 10.0.0.2

Router (config-if) # no shutdown

Router (config-if) # exit

Router (config) # interface fa0/1

Router (config-if) # ip address 20.0.0.1 255.255.255.0

Router (config-if) # ip helper-address 10.0.0.2

Router (config-if) # no shutdown

Router (config-if) # exit

Router (config-if) # exit

Step 6 : Configure PC for DHCP

for each PC : PC -> Desktop -> ip config -> DHCP

3> Output

ip adress of PC

PC0 : 10.0.0.3

PC1 : 10.0.0.4

PC2 : 20.0.0.3

PC3 : 20.0.0.5

Test 1 : within LAN

PC0 -> PC1

Ping 10.0.0.4

Status : Successful

Test 2 : Outside Lan

PC0 -> PC2

Ping 20.0.0.3

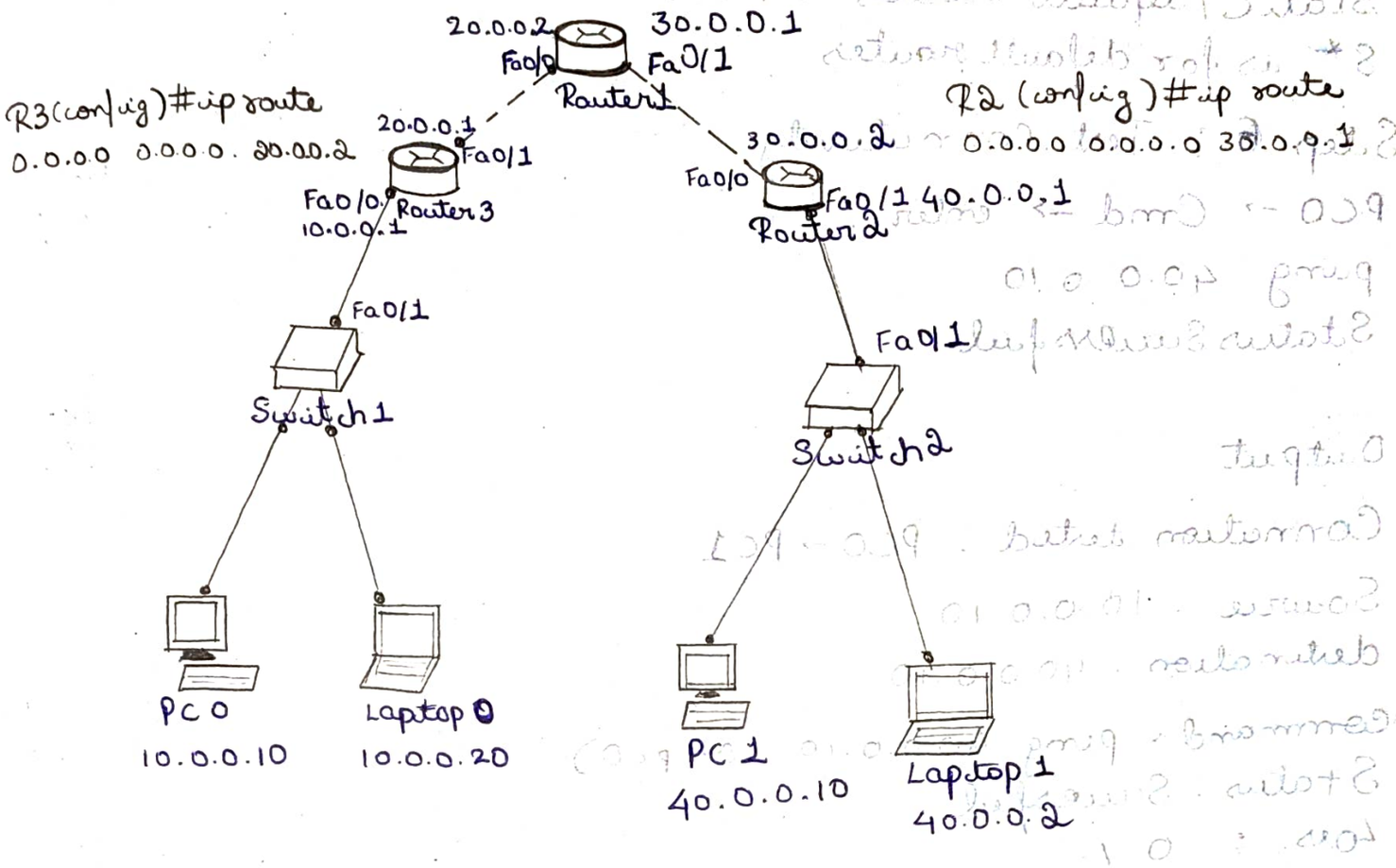
Request timed out

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How to configure static route and default route to the routers

R1(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R1(config)# ip route 40.0.0.0 255.0.0.0 30.0.0.2



Steps

1> Create above topology and configure ip of each router and PC

2> Step 2 : Configure static routes on middle router 1

Router1 > enable
> config t
> #ip route 10.0.0.0 255.0.0.0 20.0.0.1
#ip route 40.0.0.0 255.0.0.0 30.0.0.2
> #exit
> #write memory

Step 3 : Configure default route on Router 3 :
to send unknown destinations to R1, next hop is 20.0.0.2

Router 3 > enable, config t
> #ip route 0.0.0.0 0.0.0.0 20.0.0.2
> # exit
> # write memory

Step 4 : Configure on Router 2 : to send unknown to R1, next 30.0.0.1

Router 2 > enable, config t
> #ip route 0.0.0.0 0.0.0.0 30.0.0.1
> #write memory

Step 5: Check routing tables R1, R2, R3

Router 1 # show ip route

Router 2 # show ip route

Router 3 # show ip route

static / default routes can be seen

S* is for default routes

Step 6: Test connectivity

PC0 -> Cmd -> enter

ping 40.0.0.10

Status Successful

Output

Connection tested: PC0 -> PC1

Source : 10.0.0.10

destination : 40.0.0.10

Command > ping 40.0.0.10 (on pc0)

Status : Successful

Loss : 0 %

Connection tested: PC1 -> Laptop 0

Source : 40.0.0.10

destination : 10.0.0.20

Command > ping 10.0.0.20

Status : Successful

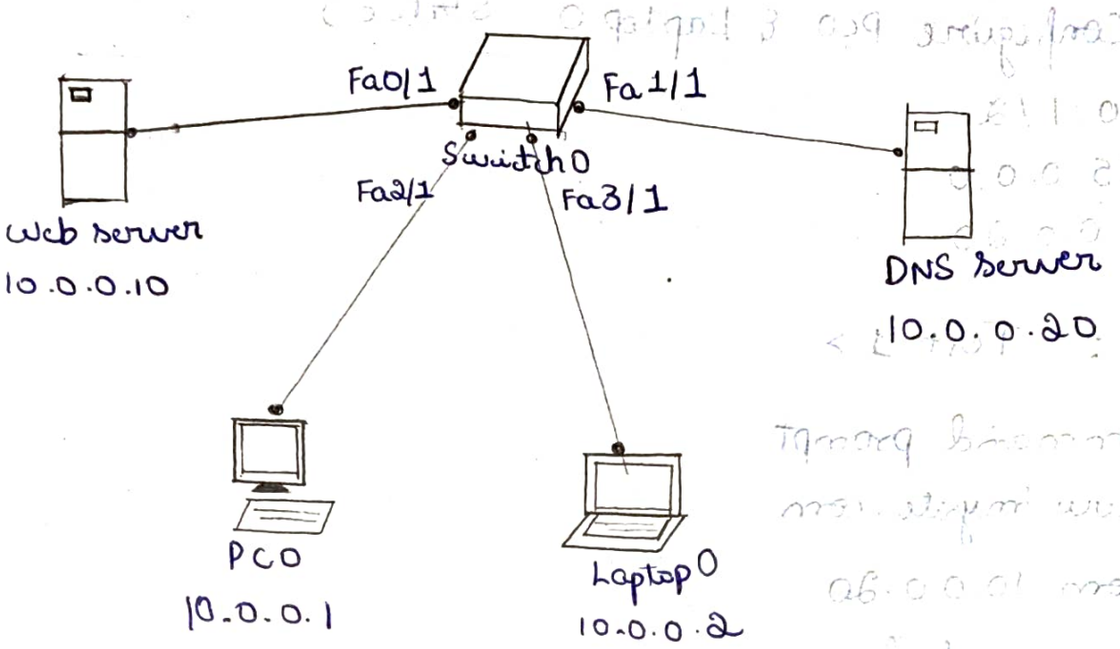
Loss : 0 %

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5> Configure Web server, DNS, within LAN.

Step 1: Create a topology like this



Step 2> Configure Web server with ip & DNS

Web Server -> Config -> fastEthernet 0

Set

IP : 10.0.0.10

Subnet : 255.0.0.0

DNS : 10.0.0.20

Step 3 : Configure DNS server with IP

dns
~~Web~~ Server -> Config -> fastethernet 0 | Config -> Global Setting

IP : 10.0.0.20

Subnet : 255.0.0

DNS : 10.0.0.20

DNS : Static

DNS Server : 10.0.0.20

Step 4 : Enable DNS sower & create a record A

DNS sower -> Services -> DNS

DNS service : ON

in resources add

Name : www.mysite.com

type : A Record

Address : 10.0.0.20

Step 5 : Enable http

~~web~~ dns Server -> Services -> HTTP

Set

Http : ON

Step 6 : Configure PC0 & Laptop 0 (Static)

ip : 10.0.0.1/2

subnet : 255.0.0.0

dns : 10.0.0.20

Output : Test 1>

PC0 -- Command prompt

ping www.mysite.com

reply from 10.0.0.20

Status : Successful

Packet loss : 0

Test 2>

PC0 -> Desktop -- Web Browser

Connection : PC0 -- Web Server

URL : www.mysite.com

Status : Successful

Result : web page displayed successfully

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